# MSc Final Exam: Decision Trees, Regression, Deep Learning and Monotonic-HMDs

Nov 2, 2026 · @Kathaj

## Instructions

This paper covers Lecture 6 (Decision Trees with ID3 and SPAM detection), Lecture 7 (Linear and Logistic Regression), the Introduction to Deep Learning lecture (MIT 15.773, Lecture 1) and the Monotonic-HMDs notes. Every question is followed by a model answer for the examiner.

| Section | Type | Questions | Marks |
|---|---|---|---|
| A | Short answer | 10 × 2 | 20 |
| B | Descriptive | 5 × 8 | 40 |
| C | Numerical problems | 5 × 8 | 40 |
| D | MCQ (optional practice / viva) | 15 × 1 | 15 |

- Total for the written paper (A + B + C): 100 marks, 3 hours.
- Show all steps in Section C. Use log base 2 for entropy. Round to 4 decimal places unless told otherwise.
- Calculators allowed.

## Section A: Short answer (10 × 2 = 20 marks)

**A1. Why is a decision tree called a rule-based classifier?**

*Answer:* Each root-to-leaf path is an IF–THEN rule, e.g. IF outlook = sunny AND humidity = high THEN play = no. A set of such rules can be drawn as a tree.

**A2. Name four types of decision-tree algorithms.**

*Answer:* ID3 (Iterative Dichotomiser 3), C4.5 (successor of ID3), CART (Classification and Regression Tree), and Random Forest (an ensemble of trees).

**A3. Define entropy and information gain.**

*Answer:* Entropy measures impurity: $H(S) = -\Sigma\, p_i \log_2 p_i$; it is 0 for a pure set and 1 for a 50/50 binary set. Information gain is the reduction in uncertainty after a split: $IG = H(S) - \Sigma\, (|S_v|/|S|)\, H(S_v)$.

**A4. How does a decision tree choose a threshold for a numerical feature?**

*Answer:* Sort the distinct values, take the midpoint between each consecutive pair as a candidate (e.g. 1, 2, 4, 5 → 1.5, 3.0, 4.5), compute IG for each, and pick the threshold with the largest IG.

**A5. Write the simple linear regression model and name each term.**

*Answer:* $y = a_0 + a_1 x + e$, where x is the input, y the observed output, $a_0$ the intercept (bias), $a_1$ the slope (weight) and e the residual error. Prediction uses $\hat{y} = a_0 + a_1 x$; the line minimises $SSE = \Sigma(y_i - \hat{y}_i)^2$.

**A6. Why is linear regression unsuitable for binary classification?**

*Answer:* A straight line can output values below 0 or above 1, which cannot be read as probabilities. Classification needs an output bounded in (0, 1), which the sigmoid provides.

**A7. Define odds and log-odds (logit).**

*Answer:* Odds = p / (1 − p), the probability of the event over the probability it does not occur. Log-odds = ln(p / (1 − p)). Logistic regression assumes the log-odds are a linear function of the features.

**A8. What is Polanyi's Paradox and why does it matter for AI?**

*Answer:* "We know more than we can tell": people do many tasks easily but cannot describe exactly how. So hand-written IF–THEN rules (traditional AI) fail to cover all situations, which motivated learning from input–output examples (machine learning).

**A9. What is a Hardware Malware Detector (HMD)?**

*Answer:* A machine-learning detector that flags malware as a computational anomaly using low-level hardware features (instruction-category counts, cache events, branches, interrupts). It is always-on, low-overhead and accurate, but can be vulnerable to adversarial evasion.

**A10. State the increasing monotonicity constraint.**

*Answer:* f is increasing monotonic if $x \leqslant x' \Rightarrow f(x) \leqslant f(x')$. For an HMD, inserting more feature-triggering instructions can never lower the malware score.

## Section B: Descriptive ($5 \times 8 = 40$ marks)

**B1. Describe the ID3 algorithm step by step. When does tree construction stop? (8)**

*Model answer:*

1. Compute the entropy of the current (whole) dataset, H(S).
2. Choose an attribute and split the dataset by it (for numerical features, by a midpoint threshold).
3. Compute the entropy of each branch.
4. Compute the information gain: IG = $H(S) - \Sigma\ (|S_v|/|S|)\ H(S_v)$.
5. Repeat steps 2–4 for every attribute.
6. Choose the attribute with the largest IG as the decision node.
7. Repeat steps 1–6 recursively on each sub-dataset.

Stop when a node is pure (a single class, H = 0), when no attribute gives a valid split, or when another stopping rule (depth, minimum samples) is reached. The leaf takes the majority class.

**B2. Explain how raw emails are converted into a numerical dataset for SPAM detection. Describe the four features used in the lecture. (8)**

*Model answer:* Pipeline: collect authorised SPAM and legitimate emails → assign ground-truth labels → parse the MIME message into header, subject, body, URLs and attachments → extract the same numerical features for every email → store one row per email with its label.

| Feature | How it is computed | Example |
|---|---|---|
| $x_1$ Suspicious keyword count | Combine subject + body, lower-case, tokenise, count words from a dictionary such as {free, winner, urgent, prize, click, offer} | "URGENT! Winner – click now for your FREE prize" → 5 |
| $x_2$ External link count | Extract hyperlinks from HTML and text, count those pointing to external domains | 2 links → 2 |
| $x_3$ Sender reputation (1–5) | From trusted history, past SPAM marks, sender vs reply-to mismatch, unknown domain | New suspicious sender → 1 |
| $x_4$ Suspicious attachment count | Inspect file names and extensions (.exe, .zip, .js) | offer.zip → 1 |

Data quality: remove duplicates, check malformed fields, use the same extraction rule for both classes, never use the SPAM-folder decision as a feature, and split into training and test sets with both classes in each.

**B3. Derive the sigmoid function from the log-odds assumption and explain how logistic regression is trained. (8)**

*Model answer:* Assume $\ln(p/(1-p)) = z$, where $z = \beta_0 + \beta_1 x$. Exponentiate: $p/(1-p) = e^z$. Then $p = e^z(1-p)$, so $p(1+e^z) = e^z$ and $p = e^z/(1+e^z)$. Dividing by $e^z$:

```
p = \sigma(z) = \frac{1}{1 + e^{-z}}
```

Training minimises the cross-entropy loss, plus an L2 penalty for a finite solution when classes are separable:

```
J_{reg}(\beta) = -\frac{1}{n}\sum_{i=1}^{n}\left[y_i \ln p_i + (1-y_i)\ln(1-p_i)\right] + \frac{\lambda}{2}\sum_{j}\beta_j^2
```

The gradient is $\partial J/\partial\beta_j = (1/n)\ \Sigma\ (p_i - y_i)\ x_{ij}$, and gradient descent updates $\beta_j \leftarrow \beta_j - \alpha\ \partial J/\partial\beta_j$. Classify as positive when $p \geqslant 0.5$. In scikit-learn, model.fit(X, y) runs this optimisation and stores model.intercept_ and model.coef_.

**B4. Trace the evolution Traditional AI → Machine Learning → Deep Learning → Generative AI. What problem did each step solve? (8)**

*Model answer:*

| Stage | Approach | Limitation it removed |
|---|---|---|
| Traditional AI (from 1956) | Experts' knowledge hand-coded as IF–THEN rules | — (success only in a few areas; Polanyi's Paradox) |
| Machine Learning | Learn input→output relationships from examples with statistics (linear/logistic regression, trees, SVM, random forests) | No need to write rules; works well on structured data |
| Deep Learning | Neural networks with many hidden layers automatically learn representations from raw data | Removed the human bottleneck of manual feature extraction for unstructured data (images, text, audio) |
| Generative AI | Deep models that produce unstructured outputs (text-to-image, image-to-text, text-to-music, LLMs, multimodal) | Earlier models could only output structured values (a number, a few probabilities) |

Deep learning's breakthrough came from three forces: new algorithmic ideas, unprecedented data from digitisation, and GPU compute.

**B5. Explain how adversarial evasion attacks defeat a regular HMD and how Monotonic-HMDs defend against them, including the reported results. (8)**

*Model answer:* A baseline linear HMD scores $y = w_1x_1 + \ldots + w_nx_n + b$ and labels malware if $y \geqslant \eta$.

- **Step 1, find vulnerable features:** in a linear model, features with negative weights (benign-favouring); in a neural network, a modified FGSM estimates useful features.
- **Step 2, perturb them:** insert instructions that raise those feature counts while keeping the malicious payload working. The score falls below $\eta$ and the malware is labelled benign.

High normal-test accuracy therefore does not mean robustness. Monotonic-HMDs build the detector only from non-vulnerable features and enforce $x \leqslant x' \Rightarrow f(x) \leqslant f(x')$, so adding instructions can never reduce the malware score.

Reported results: detection stays stable as perturbation grows (the regular HMD drops sharply); accuracy without attack falls only about 3%; overhead drops (inference time −61.11%, model size −12.01%, hardware area −53.57%, power −48.14%).

## Section C: Numerical problems ($5 \times 8 = 40$ marks)

**C1. ID3 on the weather dataset. (8)**

| Outlook | Temperature | Humidity | Windy | Play |
|---|---|---|---|---|
| sunny | hot | high | false | no |
| sunny | hot | high | true | no |
| overcast | hot | high | false | yes |
| rainy | mild | high | false | yes |
| rainy | cool | normal | false | yes |
| rainy | cool | normal | true | no |
| overcast | cool | normal | true | yes |
| sunny | mild | high | false | no |
| sunny | cool | normal | false | yes |
| rainy | mild | normal | false | yes |

| Outlook | Temperature | Humidity | Windy | Play |
|---|---|---|---|---|
| sunny | mild | normal | true | yes |
| overcast | mild | high | true | yes |
| overcast | hot | normal | false | yes |
| rainy | mild | high | true | no |

(a) Compute H(S) and the information gain of every attribute. Choose the root. (4)

(b) Complete the tree. (2)

(c) Predict play for ⟨sunny, cool, high, true⟩. (2)

*Solution:*

(a) 9 yes, 5 no: $H(S) = -(9/14)\log_2(9/14) - (5/14)\log_2(5/14) = 0.9403$.

| Attribute | Branches (yes, no) | Weighted entropy | IG |
|---|---|---|---|
| Outlook | sunny (2,3), overcast (4,0), rainy (3,2) | (5/14)(0.971) + 0 + (5/14)(0.971) = 0.6935 | **0.2467** |
| Humidity | high (3,4), normal (6,1) | (7/14)(0.9852) + (7/14)(0.5917) = 0.7885 | 0.1518 |
| Windy | false (6,2), true (3,3) | (8/14)(0.8113) + (6/14)(1.0) = 0.8922 | 0.0481 |
| Temperature | hot (2,2), mild (4,2), cool (3,1) | (4/14)(1.0) + (6/14)(0.9183) + (4/14)(0.8113) = 0.9111 | 0.0292 |

Root = **Outlook** (largest IG).

(b) Overcast is pure → yes. Sunny (2 yes, 3 no; H = 0.971): IG(Humidity) = 0.971 (high → all no, normal → all yes), IG(Temperature) = 0.571, IG(Windy) = 0.020 → split on Humidity. Rainy (3 yes, 2 no): IG(Windy) = 0.971 (false → all yes, true → all no) → split on Windy.

Rules: IF outlook = overcast THEN yes. IF sunny AND humidity = high THEN no; IF sunny AND humidity = normal THEN yes. IF rainy AND windy = true THEN no; IF rainy AND windy = false THEN yes.

(c) Outlook = sunny → humidity = high → **no** (no match).

**C2. Decision tree for SPAM detection with numerical thresholds. (8)**

| Email | $x_1$ Keywords | $x_2$ Links | $x_3$ Reputation | $x_4$ Attach. | Class |
|---|---|---|---|---|---|
| e1 | 5 | 3 | 1 | 1 | SPAM |

| Email | $x_1$ Keywords | $x_2$ Links | $x_3$ Reputation | $x_4$ Attach. | Class |
|---|---|---|---|---|---|
| e2 | 4 | 2 | 2 | 0 | SPAM |
| e3 | 6 | 4 | 1 | 1 | SPAM |
| e4 | 3 | 3 | 4 | 0 | SPAM |
| e5 | 2 | 1 | 2 | 1 | SPAM |
| e6 | 0 | 0 | 5 | 0 | Not SPAM |
| e7 | 1 | 1 | 4 | 0 | Not SPAM |
| e8 | 2 | 1 | 4 | 0 | Not SPAM |
| e9 | 3 | 2 | 5 | 0 | Not SPAM |
| e10 | 4 | 1 | 5 | 0 | Not SPAM |

(a) List the candidate thresholds for $x_3$ and compute IG for each. (3)

(b) Given best $IG(x_1) = 0.2365$, $IG(x_2) = 0.3958$ and $IG(x_4) = 0.3958$, choose the root and build the full tree. (3)

(c) Classify the unknown email $x = (5, 3, 4, 1)$. (2)

*Solution:*

(a) $H(S) = 1.0$ (5 SPAM, 5 Not SPAM). $x_3$ values {1, 2, 4, 5} → thresholds 1.5, 3.0, 4.5.

| Threshold | Left (S, N) | Right (S, N) | H_split | IG |
|---|---|---|---|---|
| $x_3 \leqslant 1.5$ | (2, 0), H = 0 | (3, 5), H = 0.9544 | 0.8 × 0.9544 = 0.7635 | 0.2365 |
| $x_3 \leqslant 3.0$ | (4, 0), H = 0 | (1, 5), H = 0.6500 | 0.6 × 0.6500 = 0.3900 | **0.6100** |
| $x_3 \leqslant 4.5$ | (5, 2), H = 0.8631 | (0, 3), H = 0 | 0.7 × 0.8631 = 0.6042 | 0.3958 |

(b) $IG(x_3 \leqslant 3.0) = 0.61$ is the largest, so the root is **$x_3 \leqslant 3.0$**. Left {e1, e2, e3, e5} is all SPAM → leaf. Right {e4, e6–e10} has 1 SPAM, 5 Not SPAM ($H = 0.65$). Inside it, $x_2 \leqslant 2.5$ gives {e6–e10} all Not SPAM and {e4} SPAM, so IG = 0.65 ($x_1$ and $x_3$ give 0.1909; $x_4$ has no valid split). Final tree: $x_3 \leqslant 3.0$ → SPAM; else $x_2 \leqslant 2.5$ → Not SPAM; else SPAM.

(c) $x_3 = 4 > 3.0$ → go to $x_2$ node; $x_2 = 3 > 2.5$ → **SPAM**.

**C3. Linear and logistic regression. (8)**

(a) Weekly sales (thousands) are $x = 1, 2, 3, 4, 5, 6$ and $y = 3, 5, 6, 8, 9, 12$. Fit $\hat{y} = a_0 + a_1 x$ and predict week 8. (4)

(b) A trained SPAM model is $z = -0.109 + 0.830x_1 + 0.368x_2 - 0.846x_3 + 0.132x_4$. Compute P(SPAM) for emails A = (5, 2, 1, 1) and B = (0, 1, 5, 0) with threshold 0.5. (3)

(c) Interpret the sign of $\beta_3$. (1)

*Solution:*

(a) $n = 6$, $\Sigma x = 21$, $\Sigma y = 43$, $\Sigma xy = 180$, $\Sigma x^2 = 91$, $\bar{x} = 3.5$, $\bar{y} = 7.1667$.

$a_1 = (n\Sigma xy - \Sigma x\Sigma y)/(n\Sigma x^2 - (\Sigma x)^2) = (1080 - 903)/(546 - 441) = 177/105 =$ **1.6857**.

$a_0 = \bar{y} - a_1\bar{x} = 7.1667 - 1.6857 \times 3.5 =$ **1.2667**. So $\hat{y} = 1.2667 + 1.6857x$, and $\hat{y}(8) = 1.2667 + 13.4857 =$ **14.75 thousand**.

(b) A: $z = -0.109 + 4.150 + 0.736 - 0.846 + 0.132 = 4.063$; $p = 1/(1 + e^{-4.063}) = 0.983 \rightarrow$ **SPAM**. B: $z = -0.109 + 0 + 0.368 - 4.230 + 0 = -3.971$; $p = 0.0185 \rightarrow$ **Not SPAM**.

(c) $\beta_3 = -0.846 < 0$: higher sender reputation lowers the log-odds of SPAM.

**C4. Neural network forward pass. (8)**

A network has 2 inputs, one hidden layer of 3 ReLU neurons and 1 sigmoid output. Trained parameters:

- $a_1 = \max(0, -0.3 + 0.5x_1 + 0.1x_2)$
- $a_2 = \max(0, 0.2 - 0.1x_1 + 0.3x_2)$
- $a_3 = \max(0, 0.5 + 0.2x_1 - 0.1x_2)$
- $y = \sigma(0.05 - 0.2a_1 - 0.3a_2 - 0.15a_3)$

(a) How many parameters does the network have? (2)

(b) Compute y for $x = (2.3, 10.2)$ and for $x = (1, 2)$. (4)

(c) The logistic regression model $P = \sigma(0.4 + 0.2 \cdot GPA + 0.5 \cdot Experience)$ is also a network. How many hidden layers does it have? Compute P for GPA = 3.8, Experience = 1.2. (2)

*Solution:*

(a) Weights: $2 \times 3 + 3 \times 1 = 9$; biases: $3 + 1 = 4$; total **13**.

(b) $x = (2.3, 10.2)$: $a_1 = \max(0, -0.3 + 1.15 + 1.02) = 1.87$; $a_2 = \max(0, 0.2 - 0.23 + 3.06) = 3.03$; $a_3 = \max(0, 0.5 + 0.46 - 1.02) = 0$. $z = 0.05 - 0.374 - 0.909 - 0 = -1.233$, $y =$ **0.226**.

$x = (1, 2)$: $a_1 = 0.4$, $a_2 = 0.7$, $a_3 = 0.5$. $z = 0.05 - 0.08 - 0.21 - 0.075 = -0.315$, $y =$ **0.4219**.

(c) Zero hidden layers (inputs go straight to the sigmoid output). $z = 0.4 + 0.76 + 0.6 = 1.76$, $P =$ **0.853**.

**C5. Evading a linear HMD and the monotonic defence. (8)**

A linear HMD uses features x = (CPU, Memory, Cache) with weights $w = (0.8, -0.5, 0.3)$, bias $b = -2$ and threshold $\eta = 0$ ($y \geqslant 0 \Rightarrow$ malware). A malware sample has $x = (5, 2, 3)$.

(a) Classify the sample. (2)

(b) Which feature is vulnerable? What is the minimum integer increase in that feature that evades detection? (3)

(c) How would a Monotonic-HMD prevent this? (3)

*Solution:*

(a) $y = 0.8(5) - 0.5(2) + 0.3(3) - 2 = 4 - 1 + 0.9 - 2 = 1.9 \geqslant 0 \rightarrow$ **malware**.

(b) Memory has a negative weight, so it is vulnerable. Adding k memory instructions gives $y = 1.9 - 0.5k < 0 \Rightarrow k > 3.8$, so **k = 4** (Memory = 6, y = −0.1 → benign). The payload is unchanged.

(c) Enforce $x \leqslant x' \Rightarrow f(x) \leqslant f(x')$: build the model only from non-vulnerable features (or constrain all weights $\geqslant 0$). Then $y = 0.8 \cdot CPU + 0.3 \cdot Cache + b$, and inserting any instructions can only keep or raise the score, so the sample stays malware.

## Section D: MCQs ($15 \times 1 = 15$ marks)

1. The entropy of a dataset with 4 F and 5 M samples is approximately: (a) 0.5 (b) 0.7219 (c) 0.9911 (d) 1.0
2. In ID3, the attribute chosen for a decision node is the one with the: (a) lowest IG (b) largest IG (c) most distinct values (d) highest entropy
3. For the 9-person dataset, Gain(Weight $\leqslant$ 160) = 0.59, Gain(Hair $\leqslant$ 5) = 0.0911, Gain(Age $\leqslant$ 40) = 0.0183. The root is: (a) Hair (b) Age (c) Weight (d) any of them
4. The candidate thresholds for distinct values 1, 2, 4, 5 are: (a) 1, 2, 4 (b) 1.5, 3.0, 4.5 (c) 2, 3, 4 (d) 3.0 only
5. C4.5 is: (a) an ensemble of trees (b) the successor of ID3 (c) a regression-only tree (d) a neural network
6. For sales data with $\hat{y} = 0.54 + 0.66x$, the predicted sales in week 12 are: (a) 5.16 (b) 7.92 (c) 8.46 (d) 9.00
7. The sigmoid output for z = 0 is: (a) 0 (b) 0.5 (c) 1 (d) e
8. If p = 0.8, the odds are: (a) 0.8 (b) 1.25 (c) 4 (d) 0.2
9. Unregularised logistic regression on perfectly separable data: (a) converges to small weights (b) has weights that keep growing (c) cannot be trained (d) becomes linear regression
10. In scikit-learn, the learned coefficients are stored in: (a) model.weights (b) model.coef_ (c) model.params (d) model.beta
11. The "human bottleneck" that deep learning removed was: (a) labelling data (b) manual feature extraction for unstructured data (c) buying GPUs (d) writing code
12. A layer in which every neuron connects to every neuron in the next layer is called: (a) sparse (b) convolutional (c) dense / fully connected (d) recurrent
13. ReLU is defined as: (a) $1/(1 + e^{-x})$ (b) $\max(0, x)$ (c) $x$ (d) $\tanh(x)$

14. In a linear HMD, which features are most vulnerable to evasion? (a) positive-weight features (b) negative-weight features (c) zero-weight features (d) the bias
15. Compared with a regular HMD, Monotonic-HMDs reduce inference time by about: (a) 12% (b) 48% (c) 54% (d) 61%

## Answer key

| Q | Answer | Q | Answer | Q | Answer |
|---|---|---|---|---|---|
| 1 | c | 6 | c | 11 | b |
| 2 | b | 7 | b | 12 | c |
| 3 | c | 8 | c | 13 | b |
| 4 | b | 9 | b | 14 | b |
| 5 | b | 10 | b | 15 | d |